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Nitrous oxide-derived vinyldiazotates: structure and reactivity

Christeena Mathew, Alexandre Genoux, Damien W. Chen, Abdusalom A. Suleymanov, Farzaneh Fadaei-Tirani, Rosario Scopelliti and Kay Severin*

Reactions of organoalkali metal compounds with N_2O are generally believed to proceed *via* diazotate intermediates. Herein, we provide experimental evidence for the formation of diazotates in reactions of $Ph_2C{=}CPhM$ (M = Li or K) with N_2O, including the crystallographic analysis of $[Ph_2C{=}CPh(N_2O)][K{\times}(18\text{-crown-}6)]$. Upon the addition of Grignard reagents, the lithium diazotate is converted into azo olefins in synthetically useful yields.

The first report about reactions of organolithium compounds with nitrous oxide was published in 1953 by Beringer and co-workers.[1] They demonstrated that these transformations produce nitrogen-containing compounds. For example, the reaction of *n*-butyllithium with N_2O afforded a hydrazone in 24% yield, whereas phenyllithium was converted into azobenzene in 7% yield, along with several other products (Fig. 1a).[1] The formation of azobenzene in the reaction of PhLi with N_2O was confirmed by independent investigations of Meier.[2]

Müller and co-workers showed that methyllithium reacts with N_2O to give diazomethane.[3] A yield of 70% could be obtained, making this procedure interesting from a preparative perspective.

The synthesis of azo-bridged ferrocene was accomplished by combining N_2O with lithiated ferrocene (yield: 25%).[4] A related approach was used to synthesize azo-bridged ferrocene oligomers.[5]

Cui and co-workers used nitrous oxide for the synthesis of benzotriazines.[6] As starting materials, they used doubly lithiated aromatic amides. The substrate scope for these reactions was found to be broad. We have shown that lithiated 2-alkylpyridines can be converted into triazolopyridines upon reaction with N_2O.[7]

In all of the aforementioned publications, it is stated that the reactions of organolithium compounds with nitrous oxide likely proceed *via* short-lived diazotates (Fig. 1a).[1–7] However, neither the isolation nor the spectroscopic characterization of such lithium diazotates was accomplished.[8] Support for the involvement of lithium diazotates in the formation of triazolopyridines was provided by a theoretical study by Poater and co-workers.[9] Additional indirect evidence comes from reports describing well-characterized diazotates resulting from reactions of N_2O with amides,[10] phosphanides,[11] N-heterocyclic carbenes,[12] or complexes of Ca(II), Sc(III), Y(III), Sm(III), and La(III).[13,14]

Below, we provide experimental evidence for the formation of diazotates in reactions of $Ph_2C{=}CPhM$ (M = Li or K) with N_2O, including a crystallographic analysis of the crown ether adduct $[Ph_2C{=}CPh(N_2O)][K{\times}(18\text{-crown-}6)]$ (Fig. 1b). Moreover, we show that the lithium diazotate can be converted into azo olefins in synthetically useful yields (Scheme 1).

First, we investigated the reaction of (1,2,2-triphenylvinyl)-lithium with N_2O. Addition of N_2O to a solution of the

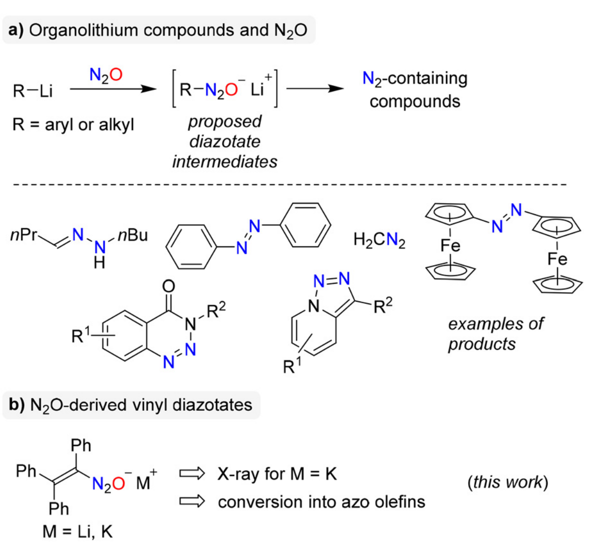
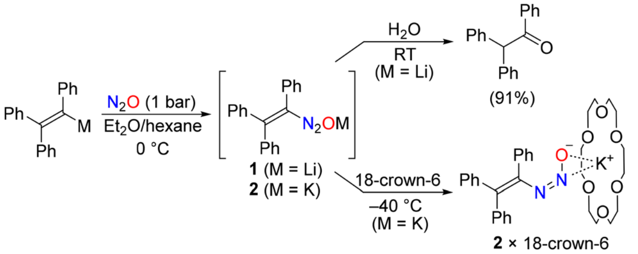


Fig. 1 (a) Reactions of organolithium compounds with nitrous oxide are proposed to involve short-lived diazotates, and examples of N_2-containing compounds formed in reactions of organolithium compounds with N_2O. (b) The N_2O-derived vinyldiazotates described in this work.

Institut des Sciences et Ingénierie Chimiques, École Polytechnique Fédérale de Lausanne (EPFL), 1015, Lausanne, Switzerland. E-mail: kay.severin@epfl.ch



Scheme 1 Synthesis and reactivity of the diazotates **1** and **2**.

vinyllithium compound in an Et_2O/hexane mixture (0.1 M) at 0 °C resulted in the immediate formation of a precipitate (**1**). We hypothesized that the precipitate **1** corresponds to a diazotate. Consistent with this assumption, treatment of the precipitate with degassed water led to the formation of the ketone 1,2,2-triphenylethan-1-one, accompanied by the liberation of N_2 (Scheme 1). Since diazotates are potentially explosive,[10] we have not tried to isolate **1** on a preparative scale.

Repeated attempts to obtain single crystals of **1** for X-ray diffraction (XRD) analysis were unsuccessful. In solution, compound **1** was unstable, suggesting that precipitation may play a beneficial role by both shifting the equilibrium toward diazotate formation and protecting it from competing side reactions. Single crystals were obtained from the corresponding potassium diazotate **2** by layering cold hexane (−40 °C) onto a solution of the potassium salt and 18-crown-6 in a THF/Et_2O mixture at −40 °C. In solution, this adduct decomposes at temperatures above −40 °C, which hampers its spectroscopic characterization and precludes accurate determination of its yield.

A single-crystal XRD analysis of **2** × 18-crown-6 confirmed that a diazotate had formed in the reaction with N_2O. Two crystallographically independent complexes were observed in the solid state, and graphic representations are depicted in Fig. 2.

In both complexes, the diazotate group adopts a *cis* configuration. Such a *cis* geometry has previously been observed for the sodium salts of methyldiazotate[15] and 4-sulfonato-phenyldiazotate,[16] as well as for the N_2O adducts of lithium amides,[10] of a potassium phosphanide,[11] and of certain N-heterocyclic carbenes.[12a,e] The diazotate groups in the two complexes show distinct geometries, with nitrogen–oxygen bond lengths of N2–O1 = 1.301(3) Å and N4–O9 = 1.277(3) Å, and nitrogen–nitrogen bond lengths of N1–N2 = 1.268(3) Å and N3–N4 = 1.238(3) Å. The variations are likely the result of the different coordination mode to the K^+ ion, with a side-on coordination for one complex (O9–K2 = 2.756(2) Å, N4–K2 = 2.815(2) Å) and a more asymmetric, O-centered coordination for the other (O1–K1 = 2.668(2) Å, N2–K1 = 2.967(2) Å).

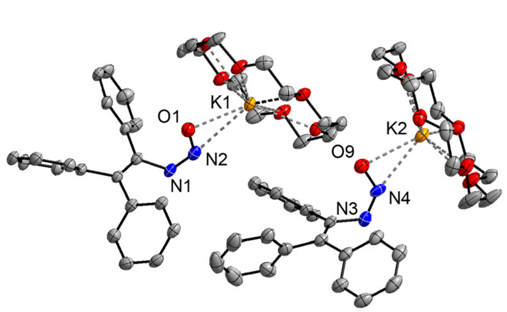
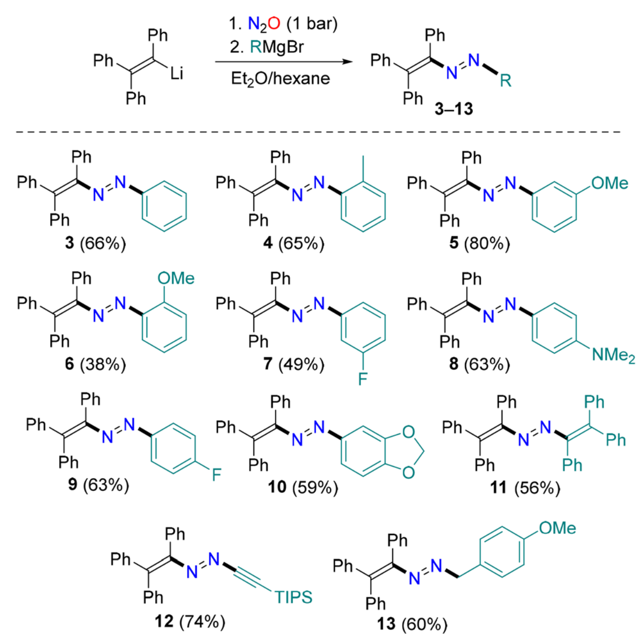


Fig. 2 Structures of the two crystallographically independent complexes in **2** × 18-crown-6 with thermal ellipsoids at 50% probability. Hydrogen atoms and co-crystallized solvent molecules (THF) are not shown.

To evaluate the bonding in the diazotate anion $[Ph_2C{=}CPh(N_2O)]^-$, we have performed a computational analysis at the B3LYP-D3(BJ)/def2-TZVP level of theory using an implicit THF solvation model. The *trans* configuration of the diazotate group was found to be more stable than the *cis* configuration, with the latter being 8.4 kJ mol^{-1} higher in Gibbs free energy. This energy difference is small, suggesting that both forms may coexist in solution. The presence of the *cis* form in crystalline **2** × 18-crown-6 can be rationalized by favorable crystal packing and/or ion pairing effects. An NBO analysis of the diazotate anion revealed that the negative charge is delocalized over the oxygen atom and the carbon-bound nitrogen atom. Accordingly, both the O–N and N–N bonds exhibit pronounced double-bond character.

The reactivity of diazotate **1** was subsequently examined (Scheme 2). In our previous studies of N_2O adducts derived from amides, we had observed that Grignard reagents can induce N–C bond formation.[10] Accordingly, we investigated the reaction of **1** with PhMgBr. We hypothesized that an analogous N–C coupling reaction could give azo olefins.[17]

Treatment of the precipitate obtained from the reaction of (1,2,2-triphenylvinyl)lithium with N_2O in an Et_2O/hexane mixture (0.1 M) with PhMgBr (1.5 equiv.) afforded the desired azo compound 1-phenyl-2-(1,2,2-triphenylvinyl)diazene (**3**) in 66% yield (Scheme 2). The driving force for the Grignard-induced



Scheme 2 Synthesis of the azo olefins **3**–**13**.

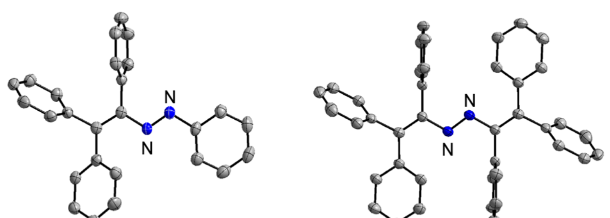
N–O bond rupture is likely the oxophilicity of Mg^{2+}. A plausible intermediate in the formation of **3** is a heteroleptic complex of type $[Ph_2C{=}CPh(N_2O)]Mg(Ph)$, which could give the azo olefin **3** along with MgO. In view of the known tendency of RLi and RMgX compounds to form complex, concentration-dependent mixtures of aggregates, we have not attempted to model the mechanism computationally.

The scope of this coupling reaction was then evaluated using a range of arylmagnesium reagents. Aryl groups bearing electron-donating or electron-withdrawing substituents were incorporated in good yields (**4**–**10**). Notably, the reaction also tolerated *ortho*-substituted arylmagnesium reagents (**4** and **6**).

Preliminary results show that the coupling chemistry can be extended to alkenyl-, alkynyl- and alkylmagnesium reagents (Scheme 2). Using conditions analogous to those employed for arylmagnesium reagents, the alkenyl-substituted azo olefin **11** was obtained in 56% yield. The alkynyl-substituted azo compound **12** was formed in 74% yield using a TIPS-substituted alkynyl Grignard reagent as coupling partner (TIPS = triisopropylsilyl). It is worth noting that, to date, there are only a few reports on the synthesis of azo alkynes.[18] Finally, we managed to obtain the *p*-methoxybenzyl-substituted azo olefin **13** in 60% yield using the corresponding alkyl Grignard reagent.

In addition to a solution-based analysis of **3**–**13** by NMR spectroscopy and high-resolution mass spectrometry, we have analyzed the solid-state structures of the azo olefins **3**, **5**, **10**, and **11** by single-crystal X-ray diffraction. Graphic representations of the structures of **3** and **11** are depicted in Fig. 3, and details about **5** and **10** can be found in the Supplementary Information (the structures are similar to that of **3**). Compound **3** shows a coplanar orientation of the 1,2-diaza-1,3-diene unit and the terminal *N*-phenyl group, pointing to electronic delocalization. For **11**, a planar 3,4-diaza-1,3,5-triene unit was observed. Both compounds display the expected *E* configuration of the azo group.

To conclude, we have demonstrated that the alkali metal compounds $Ph_2C{=}CPhM$ (M = Li or K) form diazotates when exposed to N_2O. In solution, these diazotates display limited stability. However, we managed to obtain a single crystal of the crown ether adduct $[Ph_2C{=}CPh(N_2O)][K{\times}(18\text{-crown-6})]$ at -40 °C. A crystallographic analysis revealed an N_2O adduct with a *cis* configuration of the diazotate group. This complex represents the first structurally characterized diazotate derived from an alkali metal compound and N_2O.



3 **11**

Fig. 3 Molecular structures of the azo olefins **3** and **11** as determined by single-crystal X-ray analysis. The thermal ellipsoids are at 50% probability. Hydrogen atoms are not shown.

The lithium diazotate **1** would be converted into azo olefins in reactions with Grignard reagents. This procedure represents a novel synthetic route to azo olefins, which are valuable starting materials in synthetic organic chemistry.[17] It is worth noting that compounds such as **11** or **12** would be difficult to access with the previously reported methods for synthesizing azo olefins/alkynes.[17,18] The reactivity and the physical properties of the novel azo olefins described herein remain to be explored. Interesting avenues for future investigations include the use of these compounds as photoswitches[19] or as starting material for subsequent organic transformations.[17,18]

A key aspect in the chemistry described herein is the low solubility of the diazotates **1** and **2** in the Et_2O/hexane solvent mixture, which leads to rapid precipitation. The precipitation prevents rapid follow-up reactions between the diazotates and remaining $Ph_2C{=}CPhM$, as observed in reactions of other RLi compounds with N_2O,[1–7] as well as solution-based decomposition reactions of the sensitive diazotates. This aspect should be considered in future attempts to expand the scope to vinyl diazotates other than **1** and **2** (preliminary results are described in the SI, Section S6).

Author contributions

C. M. and A. G. synthesized and characterized the new compounds, D. W. C. performed the computational analysis, A. A. S. performed important initial experiments, F. F.-T. and R. S. collected and processed the X-ray data, and C. M., A. G., and K. S. co-wrote the manuscript. All authors discussed the results and commented on the manuscript.

Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supplementary information (SI): synthetic procedures and experimental details. See DOI: https://doi.org/10.1039/d6cc05278a.

CCDC 2516496 (**5**), 2519619 (**11**), 2519620 (**3**), 2519621 (**2** × 18-C-6), and 2527917 (**10**) contain the supplementary crystallographic data for this paper.[20a–e]

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